PRANA 🫁 · SETUP GUIDE

PRANA on WhatsApp

Your autonomous AI brain, living inside the chat app you already use. Send it a message — it works with real tools and replies.

~15 min	2 min	Windows
total setup	your human steps	first-class

🤖 AUTOMATIC — PRANA / installer does it 👆 YOUR 2-MIN STEP — needs a human (account / browser)

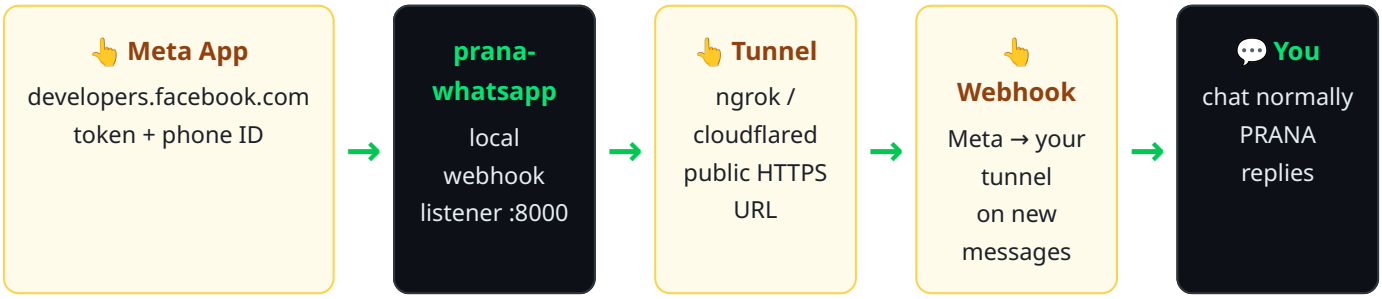
📑 Contents

1. What you're building
2. How the pieces fit (flow diagram)
3. Step 1 — Install PRANA
4. Step 2 — Save your AI key
5. Step 3 — Create the Meta developer app 👆 YOUR STEP
6. Step 4 — Copy your WhatsApp credentials 👆 YOUR STEP
7. Step 5 — Give PRANA the credentials
8. Step 6 — Start the WhatsApp listener
9. Step 7 — Open a tunnel to the internet 👆 YOUR STEP
10. Step 8 — Connect the Meta webhook 👆 YOUR STEP
11. Verify it works ✅
12. Troubleshooting
13. Quick reference

◈ What you're building

After this guide, **PRANA lives on WhatsApp**: you message it like a contact, it runs its full brain (ReAct loop, shell/files/web tools, memory), and replies in the chat. It uses the **Meta WhatsApp Business Cloud API** — official, free tier, no shady bridges, no ban risk.

◈ How the pieces fit



ℹ️ **WHY A TUNNEL?**

Meta's servers must *reach* your machine to deliver messages. Your laptop isn't publicly visible, so a tunnel (ngrok / cloudflared) gives it a temporary public HTTPS address. This is the standard pattern — nothing PRANA-specific.

1 Install PRANA

🤖 AUTOMATIC

Open **PowerShell** (or Command Prompt) and run:

PowerShell

```
$ git clone https://github.com/Thegreatprabh/Prana.git
$ cd Prana
$ python -m pip install -e .
# use `python -m pip` (not bare `pip`) — on some machines they point to different
Pythons
```

💡 **TIP**

Need Python 3.10+. Check with `python --version`. PRANA is **stdlib-only** — zero dependencies, nothing else to install.

2 Save your AI key (BYOK)

🤖 AUTOMATIC

PRANA uses **your own** API key — OpenAI, Anthropic, or Gemini. It's stored at `%USERPROFILE%\.prana\config.json` with locked-down permissions, never logged.

PowerShell

```
$ prana --save-key
Paste API key (hidden): ••••••••••••
API key saved for openai → C:\Users\YOU\.prana\config.json (0600)
```

The provider is auto-detected from the key prefix (`sk-ant-` → Anthropic, `AIza` → Gemini, `sk-` → OpenAI).

3 Create the Meta developer app

👆 YOUR 2-MIN STEP

Meta needs to know *your* app exists — this part needs a human with a browser (about 2 minutes, free):

1. Go to **developers.facebook.com → My Apps → Create App**
2. Choose type **Business** → name it (e.g. *PRANA*) → Create
3. In the app dashboard: **Add Product** → find **WhatsApp** → **Set up**

> ⚠️ **HEADS-UP**
>
> You may need to verify a phone number for the test WhatsApp Business number Meta gives you. It's part of their free sandbox — no charges.

4 Copy your WhatsApp credentials

👆 YOUR 2-MIN STEP

In your Meta app: **WhatsApp → API Setup**. Copy these two values:

What	Where	PRANA name
Temporary access token	"Temporary access token" field (copy now — it expires in 24h)	`WHATSAPP_TOKEN`
Phone number ID	"Phone number ID" under the test number	`WHATSAPP_PHONE_NUMBER_ID`

> 💡 **GOING PERMANENT**
>
> The temporary token expires in ~24 hours — fine for testing. For always-on use: **Business Settings → System Users → Add** → assign the `whatsapp_business_messaging` permission → generate a permanent token. Same variable, longer life.

5 Give PRANA the credentials

🤖 AUTOMATIC

Recommended (persistent): add a `whatsapp` section to `%USERPROFILE%\.prana\config.json`. Pick any secret string as your verify token — you'll paste the same value into Meta in Step 8.

%USERPROFILE%\.prana\config.json

```
{
  "whatsapp": {
    "token": "EAAB...paste-your-token...",
    "phone_number_id": "123456789012345",
    "verify_token": "prana-secret-123"
  }
}
```

Alternative (this terminal session only): `$env:WHATSAPP_TOKEN="..."`, `$env:WHATSAPP_PHONE_NUMBER_ID="..."`, `$env:WHATSAPP_VERIFY_TOKEN="..."` in PowerShell.

6 Start the WhatsApp listener

🤖 AUTOMATIC

PowerShell

```
$ prana-whatsapp --port 8000
💬 PRANA listening for WhatsApp on 127.0.0.1:8000/webhook — Ctrl+C to stop
Next: expose this listener publicly (ngrok/cloudflared) and set the Meta app
webhook Callback URL to https://<your-tunnel>/webhook (subscribe: messages).
```

Leave this window running. Open a **second** terminal for the next step.

7 Open a tunnel to the internet

👆 YOUR 2-MIN STEP

In the second terminal, run one of these (pick whichever you have):

PowerShell — terminal 2

```
$ ngrok http 8000
# → copy the https URL, e.g. https://a1b2-c3d4.ngrok-free.app
# — or —
$ cloudflared tunnel --url http://localhost:8000
# → copy the https URL, e.g. https://random-words.trycloudflare.com
```

> ⚠️ **FREE TUNNELS ROTATE**
>
> Free ngrok/cloudflared URLs **change every restart**. Each time the URL changes, repeat Step 8 with the new URL. (Paid ngrok gives a fixed domain.)

8 Connect the Meta webhook

👆 YOUR 2-MIN STEP

Back in **developers.facebook.com → your app → WhatsApp → Configuration**:

1. **Callback URL:** `https://<your-tunnel-url>/webhook`
2. **Verify token:** the exact string from Step 5 (`prana-secret-123`)
3. Click **Verify and Save** — Meta calls your listener; PRANA answers the handshake automatically
4. In **Webhook fields**, click **Manage** → subscribe to **`messages`**

> 💡 **WHAT SUCCESS LOOKS LIKE**
>
> Your `prana-whatsapp` terminal prints `whatsapp: webhook verified`. Meta shows a green checkmark on the webhook.

✓ Verify it works

✅ Send a WhatsApp message to the test number from your phone

✅ PRANA replies in the chat (first message may take a few seconds — it's thinking)

✅ Try `help` → command list, and a real task like *"summarize what you can do"*

✅ Conversation history is per-number — it remembers your thread

Troubleshooting

Error / symptom	Fix
`error: no WhatsApp token — set WHATSAPP_TOKEN…`	Token not found. Check the `whatsapp.token` value in config.json, or the `WHATSAPP_TOKEN` env var in the *same* terminal.
`error: no WhatsApp phone_number_id…`	Copy "Phone number ID" from Meta app → WhatsApp → API Setup (it's the numeric ID, not the phone number).
`warning: no webhook verify token set…`	Set `WHATSAPP_VERIFY_TOKEN` — without it Meta's verification rejects everything.
`Meta: webhook verification failed (403)`	Verify-token mismatch. The string in Meta's dashboard must **exactly** equal your `verify_token`.
`Messages sent, nothing arrives`	Tunnel URL changed? Update the Callback URL. Also confirm you subscribed to the `messages` webhook field, and the sender is an allowed test recipient.
`PRANA replies, then Meta retries duplicates`	Your listener must answer POSTs with 200 fast — PRANA does this by design (agent failures are logged, never retried).
`Temporary token stopped working`	It expires in ~24h. Generate a permanent system-user token (Step 4 tip).

ℹ️ HONEST META RULE

WhatsApp Business only lets *you* start conversations freely. If PRANA pings you proactively after 24h of silence, Meta requires a pre-approved message template. Replies inside an active chat are always free.

Quick reference

Item	Value
Start listener	`prana-whatsapp --port 8000`
Webhook path	`/webhook` (customize: `--webhook-path`)
Env vars	`WHATSAPP_TOKEN` · `WHATSAPP_PHONE_NUMBER_ID` · `WHATSAPP_VERIFY_TOKEN`
Config file	`%USERPROFILE%\.prana\config.json` → `"whatsapp": {...}`
Cloud API	`https://graph.facebook.com/v22.0`
Message limit	4000 chars per message (auto-chunked)
Local commands	`start` · `help` (answered instantly, zero tokens)